RISK FACTORS ASSESSMENT AND SCREENING PROCEDURES AMONG CLIENTS EXPERIENCING AN ACUTE BIOLOGIC CRISIS .

CK. CAYAGO

ACUTE BIOLOGIC CRISIS

- AN ACUTE BIOLOGIC CRISIS REFERS TO A SUDDEN, SEVERE DISRUPTION OF PHYSIOLOGICAL FUNCTIONING THAT THREATENS LIFE OR ORGAN INTEGRITY AND REQUIRES IMMEDIATE CLINICAL INTERVENTION.

ACUTE BIOLOGIC CRISIS

- **EXAMPLES INCLUDE:**
  - SEPSIS
  - ACUTE RESPIRATORY FAILURE
  - HYPOGLYCEMIA OR HYPERGLYCEMIA CRISES
  - ACUTE CARDIAC EVENTS
  - SEVERE ELECTROLYTE IMBALANCE
  - DRUG TOXICITY OR WITHDRAWAL
  - ACUTE NEUROLOGIC COMPROMISE (E.G., STROKE)

ACUTE BIOLOGIC CRISIS

BIOLOGICAL AND MEDICAL RISK FACTORS

- CHRONIC ILLNESSES (E.G., DIABETES, CARDIOVASCULAR DISEASE, RENAL FAILURE)
- IMMUNOCOMPROMISED STATES
- POLYPHARMACY OR HIGH-RISK MEDICATIONS
- RECENT SURGERY OR INVASIVE PROCEDURES
- SUBSTANCE USE OR ABRUPT WITHDRAWAL
- POOR NUTRITIONAL STATUS OR DEHYDRATION
- HISTORY OF PREVIOUS ACUTE BIOLOGIC CRISES

ACUTE BIOLOGIC CRISIS

RISK FACTORS ASSESSMENT:

- **BIOLOGICAL FACTORS**
  - AGE (ELDERLY, INFANTS)
  - CHRONIC ILLNESSES (DIABETES, HYPERTENSION, ASTHMA)
  - IMMUNOSUPPRESSION
  - PREGNANCY

ACUTE BIOLOGIC CRISIS

BEHAVIORAL AND LIFESTYLE FACTORS

- SUBSTANCE USE
- POOR MEDICATION ADHERENCE
- SMOKING
- POOR NUTRITION

ACUTE BIOLOGIC CRISIS

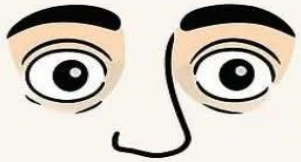
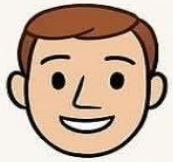
ENVIRONMENTAL AND SITUATIONAL FACTORS

- EXPOSURE TO TOXINS
- LACK OF ACCESS TO HEALTHCARE
- RECENT SURGERY OR TRAUMA

ACUTE BIOLOGIC CRISIS

- **SCREENING PROCEDURES**
  - **CLINICAL SCREENING**
    - VITAL SIGNS (BLOOD PRESSURE, PULSE, RESPIRATION, TEMPERATURE, OXYGEN SATURATION)
    - LEVEL OF CONSCIOUSNESS (E.G., GLASGOW COMA SCALE)
    - PAIN ASSESSMENT

Glasgow coma scale

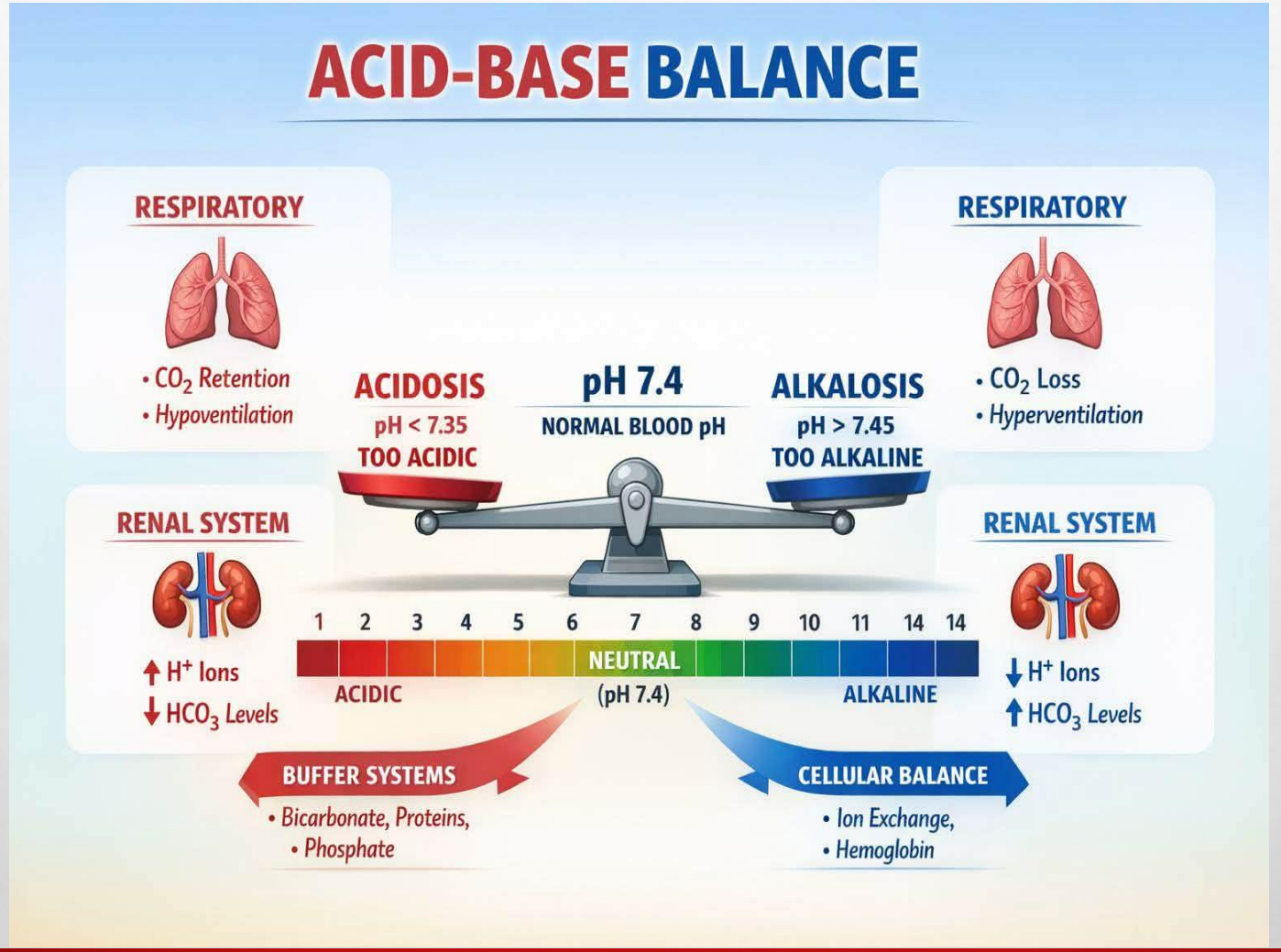
EYE OPENING		VERBAL RESPONSE		MOTOR RESPONSE	
					
Spontaneous	4	Oriented	5	Obey commands	6
To sound	3	Confused	4	Localizing	5
To pressure	2	Words	3	Normal flexion	4
None	1	Sounds	2	Abnormal flexion	3
		None	1	Extension	2

GLASGOW COMA SCALE SCORE

Mild	Moderate	Severe
13–15	9–12	3–8

ACUTE BIOLOGIC CRISIS

- **DIAGNOSTIC SCREENING:**
  - BLOOD GLUCOSE TESTING
  - ECG
  - BLOOD TESTS (CBC, ELECTROLYTES, ABGS)
  - IMAGING (X-RAY, CT SCAN) WHEN NEEDED


ACID-BASE BALANCE
RESPIRATORY
• CO_2 Retention
• Hypoventilation
RESPIRATORY
• CO_2 Loss
• Hyperventilation
ACIDOSIS
pH < 7.35
TOO ACIDIC
pH 7.4
NORMAL BLOOD pH
ALKALOSIS
pH > 7.45
TOO ALKALINE
RENAL SYSTEM
↑ H^+ Ions
↓ HCO_3 Levels
RENAL SYSTEM
↓ H^+ Ions
↑ HCO_3 Levels
1 2 3 4 5 6 7 8 9 10 11 14 14
ACIDIC
NEUTRAL
(pH 7.4)
ALKALINE
BUFFER SYSTEMS
• Bicarbonate, Proteins,
• Phosphate
CELLULAR BALANCE
• Ion Exchange,
• Hemoglobin

ACUTE BIOLOGIC CRISIS

- **RISK STRATIFICATION TOOLS**
  - EARLY WARNING SCORES (E.G., MEWS, NEWS)
  - SEPSIS SCREENING TOOLS
  - STROKE SCALES (FAST, NIHSS)

National Institutes of Health Stroke Scale (NIHSS)

The NIHSS is a test used to check how serious a stroke is. The test checks things like:



Awareness



Consciousness



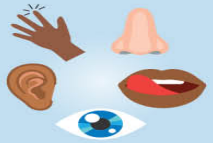
Coordination



Language and speech



Movement



Senses

A higher score means the stroke is more serious.

0 to 5	Minor stroke
6 to 15	Moderate stroke
16 to 20	Moderate to severe stroke
21 to 42	Severe stroke

A normal NIHSS score is 0.
A 42 may mean you're in a coma. The exact symptoms you have at each severity level can vary.

Cleveland Clinic

National Early Warning Score (NEWS)*

PHYSIOLOGICAL PARAMETERS	3	2	1	0	1	2	3
Respiration Rate	≤8		9 - 11	12 - 20		21 - 24	≥25
Oxygen Saturations	≤91	92 - 93	94 - 95	≥96			
Any Supplemental Oxygen		Yes		No			
Temperature	≤35.0		35.1 - 36.0	36.1 - 38.0	38.1 - 39.0	≥39.1	
Systolic BP	≤90	91 - 100	101 - 110	111 - 219			≥220
Heart Rate	≤40		41 - 50	51 - 90	91 - 110	111 - 130	≥131
Level of				A			V, P or U

Modified Early Warning System (MEWS)

	3	2	1	0	1	2	3
Respiratory rate per minute		Less than 8	8	9–17	18–20	21–29	greater than or = to 30
Heart rate per minute		Less than 40	40–50	51–100	101–110	111–129	greater than or = to 130
Systolic blood pressure	Less than or = to 70	71–80	81–100	101–159	160–199	200–220	greater than 220
Conscious level (AVPU)	Unrespon-sive	Responds to pain	Responds to voice	Alert	Agitation or confusion	New on set of agitation or confusion	
Temperature		Less than 95.0°F (35.0°C)	95.0–96.8° F (35.05–36° C)	96.9–100.4°F (36.05–38° C)	100.5–101.3° F (38.05–38.5° C)	greater than or = to 101.4°F (38.55°C)	

Source: Kathy D.Duncan RN; Christine McMullan MPA; Barbara M. Mills DNP, PNP, RN, ACNPC, ANPC, CCRN, PCCN – Nursing February 2012

Green = 0–2 Score

Yellow = 3 Score

Orange = 4–5 Score

Red = greater than 6 Score

ACUTE BIOLOGIC CRISIS

- **PURPOSE OF RISK ASSESSMENT AND SCREENING**
  - TO **IDENTIFY LIFE-THREATENING CONDITIONS EARLY**
  - TO **PRIORITIZE CARE** AND ALLOCATE RESOURCES
  - TO **PREVENT COMPLICATIONS OR DEATH**
  - TO **GUIDE IMMEDIATE TREATMENT AND REFERRAL**

- **“PRINCIPLES AND TECHNIQUES OF PHYSICAL EXAMINATION (ACUTE BIOLOGIC CRISIS OR LIFE-THREATENING SITUATION)”**

  REFERS TO THE **SYSTEMATIC, RAPID, AND PRIORITIZED APPROACH USED BY HEALTHCARE PROFESSIONALS TO ASSESS A PATIENT WHO IS CRITICALLY ILL OR AT IMMEDIATE RISK OF DEATH**, TO IDENTIFY LIFE-THREATENING PROBLEMS AND INITIATE URGENT INTERVENTION**.**

1. LIFE-SAVING FIRST

- FOCUS ON CONDITIONS THAT THREATEN LIFE BEFORE ANYTHING ELSE.

2. SYSTEMATIC AND RAPID

- FOLLOW AN ORGANIZED APPROACH TO AVOID MISSING CRITICAL PROBLEMS.

3. ABC PRIORITY

- **A – AIRWAY**
- **B – BREATHING**
- **C – CIRCULATION**
- (OFTEN EXTENDED TO **D – DISABILITY** AND **E – EXPOSURE**)

4. MINIMAL BUT FOCUSED

- ONLY ESSENTIAL EXAMINATIONS ARE DONE INITIALLY TO AVOID DELAYS.

5. CONTINUOUS REASSESSMENT

- PATIENT IS RE-EXAMINED AFTER EVERY INTERVENTION.

6. DO NO HARM

- AVOID UNNECESSARY MOVEMENT OR PROCEDURES THAT COULD WORSEN THE CONDITION.

Adult Out-of-Hospital Chain of Survival



Adult In-Hospital Chain of Survival



Pediatric Out-of-Hospital Chain of Survival



Pediatric In-Hospital Chain of Survival



TECHNIQUES OF PHYSICAL EXAMINATION IN LIFE-THREATENING SITUATIONS

- **1. INSPECTION**
  - OBSERVE SKIN COLOR (PALLOR, CYANOSIS)
  - CHEST MOVEMENT AND BREATHING PATTERN
  - LEVEL OF CONSCIOUSNESS
  - SIGNS OF BLEEDING OR TRAUMA

- **2. PALPATION**
  - CHECK PULSE STRENGTH AND RATE
  - ASSESS TEMPERATURE AND SKIN MOISTURE
  - IDENTIFY TENDERNESS, SWELLING, OR DEFORMITY

- **3. PERCUSSION *(USED SELECTIVELY)***
  - DETECT FLUID OR AIR IN LUNGS OR ABDOMEN WHEN APPROPRIATE

- **4. AUSCULTATION**
  - LUNG SOUNDS (WHEEZES, CRACKLES, ABSENT BREATH SOUNDS)
  - HEART SOUNDS (RATE, RHYTHM, MURMURS)
  - BOWEL SOUNDS (IF INDICATED)

Component	Focus of Examination
A – Airway	Patency, obstruction, cervical spine protection
B – Breathing	Rate, depth, oxygen saturation, chest symmetry
C – Circulation	Pulse, blood pressure, bleeding, capillary refill
D – Disability	Disability Neurological status (GCS, pupil reaction)
E – Exposure	Full body check while preventing hypothermia

AIRWAY (A)

- **SITUATION:** UNCONSCIOUS TRAUMA PATIENT
- **NURSING EXAMPLE:**
  - NURSE INSPECTS FOR AIRWAY OBSTRUCTION (VOMIT, BLOOD).
  - PERFORMS HEAD-TILT–CHIN-LIFT OR JAW-THRUST (IF SPINAL INJURY SUSPECTED).
  - SUCTION SECRETIONS AND APPLIES OXYGEN.
- **MEDICAL EXAMPLE:**
  - PHYSICIAN INSERTS AN OROPHARYNGEAL AIRWAY OR PERFORMS ENDOTRACHEAL INTUBATION IF THE AIRWAY IS COMPROMISED.

BREATHING (B)

- **SITUATION:** PATIENT WITH SEVERE ASTHMA ATTACK
- **NURSING EXAMPLE:**
  - INSPECTS CHEST MOVEMENT AND COUNTS RESPIRATORY RATE.
  - AUSCULTATES LUNGS AND NOTES WHEEZING OR ABSENT BREATH SOUNDS.
  - MONITORS OXYGEN SATURATION AND ADMINISTERS NEBULIZED BRONCHODILATORS.
- **MEDICAL EXAMPLE:**
  - DOCTOR ORDERS ARTERIAL BLOOD GASES AND PRESCRIBES IV CORTICOSTEROIDS OR MECHANICAL VENTILATION IF NEEDED.

CIRCULATION (C)

- **SITUATION:** PATIENT WITH MASSIVE BLEEDING AFTER AN ACCIDENT
- **NURSING EXAMPLE:**
  - PALPATES PULSE (WEAK, RAPID).
  - CHECKS BLOOD PRESSURE AND CAPILLARY REFILL.
  - APPLIES DIRECT PRESSURE TO CONTROL BLEEDING AND STARTS IV FLUIDS.
- **MEDICAL EXAMPLE:**
  - PHYSICIAN ORDERS BLOOD TRANSFUSION AND EMERGENCY SURGERY IF HEMORRHAGE PERSISTS.

DISABILITY (D) – NEUROLOGICAL STATUS

- **SITUATION:** SUSPECTED STROKE PATIENT
- **NURSING EXAMPLE:**
  - ASSESSES LEVEL OF CONSCIOUSNESS USING THE GLASGOW COMA SCALE (GCS).
  - CHECKS PUPIL SIZE AND REACTION TO LIGHT.
  - PERFORMS FAST ASSESSMENT (FACE DROOP, ARM WEAKNESS, SPEECH DIFFICULTY).
- **MEDICAL EXAMPLE:**
  - DOCTOR ORDERS AN URGENT CT SCAN AND EVALUATES ELIGIBILITY FOR THROMBOLYTIC THERAPY.

- **EXPOSURE (E)**
  - **SITUATION:** SEPTIC PATIENT WITH HIGH FEVER
  - **NURSING EXAMPLE:**
    - EXPOSES THE PATIENT TO ASSESS FOR RASHES, WOUNDS, OR INFECTION SITES.
    - MONITORS BODY TEMPERATURE AND PREVENTS HYPOTHERMIA WITH WARM BLANKETS.
  - **MEDICAL EXAMPLE:**
    - PHYSICIAN IDENTIFIES INFECTION SOURCE AND ORDERS BROAD-SPECTRUM ANTIBIOTICS.

- **TECHNIQUE-BASED EXAMPLES**
  - **INSPECTION**
    - **NURSING:** OBSERVES CYANOSIS IN THE LIPS, INDICATING HYPOXIA.
    - **MEDICAL:** NOTES ASYMMETRIC CHEST MOVEMENT SUGGESTING PNEUMOTHORAX.
  - **PALPATION**
    - **NURSING:** FEELS COLD, CLAMMY SKIN IN SHOCK.
    - **MEDICAL:** DETECTS ABDOMINAL RIGIDITY SUGGESTING INTERNAL BLEEDING.
  - **AUSCULTATION**
    - **NURSING:** HEARS CRACKLES IN THE LUNGS OF A HEART FAILURE PATIENT.
    - **MEDICAL:** IDENTIFIES ABSENT BOWEL SOUNDS IN THE ACUTE ABDOMEN.

- **PURPOSE OF THESE PRINCIPLES AND TECHNIQUES**
  - RAPID IDENTIFICATION OF LIFE-THREATENING CONDITIONS
  - IMMEDIATE INITIATION OF TREATMENT
  - PREVENTION OF DETERIORATION AND DEATH
  - GUIDANCE FOR FURTHER DIAGNOSTIC TESTING

RESULTS AND IMPLICATIONS OF DIAGNOSTIC/LABORATORY EXAMINATIONS

- COMPLETE BLOOD COUNT (CBC)
- BLOOD GLUCOSE TESTING
- ARTERIAL BLOOD GASES (ABG)
- ELECTROLYTE PANEL
- CARDIAC ENZYMES (TROPONIN, CK-MB)
- COAGULATION PROFILE (PT, INR, APTT)
- IMAGING STUDIES (X-RAY, CT SCAN, MRI)

PT APTT INR
Coagulation Test Comparison

@Medical Laboratory Scientist - MLS

PT
Prothrombin Time



Evaluates Extrinsic & Common Pathways

- Warfarin Therapy
- Liver Function
- Vit. K Deficiency



Reference Range:
11–13 Seconds

APTT
Activated Partial Thromboplastin Time



Evaluates Intrinsic & Common Pathways

- Heparin Therapy
- Hemophilia
- Bleeding Disorders



Reference Range:
25–35 Seconds

INR
International Normalized Ratio



Standardized PT Calculation

- Warfarin Monitoring
- Target Range: **2.0–3.0**
- Mechanical Valves: **2.5–3.5**



Target Range:
2.0–3.0

RESULTS AND IMPLICATIONS OF DIAGNOSTIC/LABORATORY EXAMINATIONS

- **CENTRAL VENOUS PRESSURE (CVP)**
  - **DEFINITION:**
    - CVP MEASURES THE **PRESSURE IN THE RIGHT ATRIUM** AND REFLECTS **RIGHT VENTRICULAR PRELOAD** AND OVERALL **FLUID VOLUME STATUS**.

- **NORMAL VALUE**
  - **2–6 MMHG** (OR **5–10 CM H_2O**)
    - **INTERPRETATION OF RESULTS**
      - **LOW CVP**
        - HYPOVOLEMIA (DEHYDRATION, HEMORRHAGE)
        - SEPTIC SHOCK (EARLY PHASE)
      - **HIGH CVP**
        - FLUID OVERLOAD
        - RIGHT-SIDED HEART FAILURE
        - PULMONARY HYPERTENSION
        - CARDIAC TAMPONADE

- **PULMONARY CAPILLARY WEDGE PRESSURE (PCWP)**
  - **DEFINITION**
  - PCWP ESTIMATES **LEFT ATRIAL PRESSURE** AND REFLECTS **LEFT VENTRICULAR PRELOAD**.
- **NORMAL VALUE**
  - **6–12 MMHG**
- **INTERPRETATION OF RESULTS**
  - **LOW PCWP**
    - HYPOVOLEMIA
    - DEHYDRATION
  - **HIGH PCWP**
    - LEFT-SIDED HEART FAILURE
    - PULMONARY EDEMA
    - MITRAL VALVE DISEASE
    - FLUID OVERLOAD

1. ACUTE PULMONARY FAILURE

- **DEFINITION**
  - ACUTE PULMONARY FAILURE IS A **SUDDEN INABILITY OF THE LUNGS TO PROVIDE ADEQUATE OXYGENATION AND/OR REMOVE CARBON DIOXIDE**, RESULTING IN **HYPOXEMIA, HYPERCAPNIA, OR BOTH**, AND POSING AN IMMEDIATE THREAT TO LIFE.

1. IMPAIRED GAS EXCHANGE

- DAMAGE TO THE ALVEOLAR–CAPILLARY MEMBRANE
- FLUID, PUS, OR BLOOD FILLING ALVEOLI

RESULT:

- ↓ OXYGEN DIFFUSION INTO BLOOD
- ↑ CARBON DIOXIDE RETENTION

EXAMPLES:

- ACUTE RESPIRATORY DISTRESS SYNDROME (ARDS)
- PNEUMONIA
- PULMONARY EDEMA

2. VENTILATION–PERFUSION (V/Q) MISMATCH

- AREAS OF THE LUNG ARE VENTILATED BUT NOT PERFUSED, OR PERFUSED BUT NOT VENTILATED

RESULT:

- INADEQUATE OXYGEN DELIVERY DESPITE BREATHING

EXAMPLES:

- PULMONARY EMBOLISM
- ATELECTASIS
- SEVERE COPD EXACERBATION

3. ALVEOLAR COLLAPSE (ATELECTASIS)

- LOSS OF SURFACTANT OR AIRWAY OBSTRUCTION
- REDUCED LUNG COMPLIANCE

- **RESULT:**
  - DECREASED FUNCTIONAL LUNG VOLUME
  - INCREASED WORK OF BREATHING
- **EXAMPLES:**
  - POST-OPERATIVE PATIENTS
  - PROLONGED BED REST
  - MUCUS PLUGGING

4. AIRWAY OBSTRUCTION

- NARROWED OR BLOCKED AIRWAYS

- **RESULT:**
  - REDUCED AIRFLOW
  - CO_2 RETENTION (HYPERCAPNIA)
- **EXAMPLES:**
  - ACUTE ASTHMA ATTACK
  - ANAPHYLAXIS
  - FOREIGN BODY ASPIRATION

- **5. RESPIRATORY MUSCLE FATIGUE OR FAILURE**
  - INABILITY OF THE RESPIRATORY MUSCLES TO SUSTAIN VENTILATION
  - **RESULT:**
    - HYPOVENTILATION
    - RESPIRATORY ACIDOSIS
  - **EXAMPLES:**
    - NEUROMUSCULAR DISORDERS
    - SEVERE SEPSIS
    - PROLONGED MECHANICAL VENTILATION

- **PHYSIOLOGIC CONSEQUENCES:**
  - **HYPOXEMIA** → TISSUE HYPOXIA → ORGAN DYSFUNCTION
  - **HYPERCAPNIA** → RESPIRATORY ACIDOSIS
  - **INCREASED PULMONARY PRESSURES**
  - **PROGRESSION TO RESPIRATORY ARREST IF UNTREATED**

- **CLINICAL MANIFESTATIONS:**
  - DYSPNEA AND TACHYPNEA
  - CYANOSIS
  - USE OF ACCESSORY MUSCLES
  - DECREASED OXYGEN SATURATION
  - ALTERED MENTAL STATUS

- **NURSING IMPLICATIONS:**
  - MONITOR OXYGEN SATURATION AND RESPIRATORY RATE
  - POSITION THE PATIENT TO MAXIMIZE LUNG EXPANSION
  - ADMINISTER OXYGEN THERAPY
  - PERFORM SUCTIONING AND AIRWAY CLEARANCE
  - PREPARE FOR MECHANICAL VENTILATION IF INDICATED

2. MASSIVE BLEEDING

- **DEFINITION**
  - MASSIVE BLEEDING (HEMORRHAGE) IS A **RAPID AND SEVERE LOSS OF BLOOD VOLUME** THAT LEADS TO **DECREASED TISSUE PERFUSION, HYPOVOLEMIC SHOCK, AND POTENTIAL MULTI-ORGAN FAILURE** IF NOT PROMPTLY CONTROLLED.

1. ACUTE LOSS OF INTRAVASCULAR VOLUME

- SUDDEN LOSS OF WHOLE BLOOD OR PLASMA

RESULT:

- ↓ VENOUS RETURN
- ↓ CARDIAC OUTPUT
- ↓ BLOOD PRESSURE

EXAMPLES:

- TRAUMA
- GASTROINTESTINAL BLEEDING
- POST-OPERATIVE HEMORRHAGE
- OBSTETRIC HEMORRHAGE

2. DECREASED OXYGEN DELIVERY TO TISSUES

- LOSS OF RED BLOOD CELLS REDUCES OXYGEN-CARRYING CAPACITY

RESULT:

- TISSUE HYPOXIA
- SHIFT TO ANAEROBIC METABOLISM
- LACTIC ACIDOSIS

3. COMPENSATORY MECHANISMS

- SYMPATHETIC NERVOUS SYSTEM ACTIVATION
- RELEASE OF CATECHOLAMINES

EFFECTS:

- TACHYCARDIA
- VASOCONSTRICTION
- INCREASED RESPIRATORY RATE
- (INITIALLY MAINTAINS PERFUSION, BUT FAILS IF BLEEDING CONTINUES)

- **4. COAGULOPATHY DEVELOPMENT**
  - LOSS AND DILUTION OF CLOTTING FACTORS
  - CONSUMPTION OF PLATELETS

RESULT:

  - IMPAIRED CLOT FORMATION
  - ONGOING OR WORSENING BLEEDING

(“LETHAL TRIAD”: HYPOTHERMIA, ACIDOSIS, COAGULOPATHY)

5. PROGRESSION TO HYPOVOLEMIC SHOCK

- INADEQUATE CIRCULATION TO VITAL ORGANS

RESULT:

- RENAL FAILURE
- ALTERED MENTAL STATUS
- CARDIAC ARREST IF UNTREATED

- **CLINICAL MANIFESTATIONS:**
  - HYPOTENSION
  - TACHYCARDIA
  - PALE, COLD, CLAMMY SKIN
  - DECREASED URINE OUTPUT
  - ALTERED LEVEL OF CONSCIOUSNESS

- **NURSING IMPLICATIONS:**
  - RAPID ASSESSMENT OF BLEEDING SOURCE
  - MONITOR VITAL SIGNS AND LEVEL OF CONSCIOUSNESS
  - ESTABLISH LARGE-BORE IV ACCESS
  - ADMINISTER IV FLUIDS AND BLOOD PRODUCTS AS ORDERED
  - MONITOR INTAKE AND OUTPUT
  - PREVENT HYPOTHERMIA

3. EXTENSIVE SURGERIES

- **DEFINITION**
  - EXTENSIVE SURGERIES ARE **MAJOR, PROLONGED, OR HIGHLY INVASIVE SURGICAL PROCEDURES** THAT SIGNIFICANTLY DISRUPT NORMAL BODY SYSTEMS, LEADING TO **PHYSIOLOGIC STRESS, ALTERED PERFUSION, INFLAMMATORY RESPONSES, AND RISK OF LIFE-THREATENING COMPLICATIONS**.

1. TISSUE INJURY AND INFLAMMATORY RESPONSE

- SURGICAL INCISION AND MANIPULATION CAUSE TISSUE DAMAGE
- RELEASE OF INFLAMMATORY MEDIATORS (CYTOKINES, PROSTAGLANDINS)

RESULT:

- INCREASED CAPILLARY PERMEABILITY
- FLUID SHIFT INTO INTERSTITIAL SPACES
- EDEMA AND DECREASED CIRCULATING VOLUME

2. BLOOD AND FLUID LOSS

- INTRAOPERATIVE BLOOD LOSS
- THIRD-SPACE FLUID SHIFTS

RESULT:

- HYPOVOLEMIA
- ↓ CARDIAC OUTPUT
- RISK OF HYPOVOLEMIC SHOCK

3. STRESS RESPONSE TO SURGERY

- ACTIVATION OF THE SYMPATHETIC NERVOUS SYSTEM AND ENDOCRINE SYSTEM
- RELEASE OF CATECHOLAMINES, CORTISOL, AND GLUCAGON

RESULT:

- TACHYCARDIA AND HYPERTENSION (EARLY)
- HYPERGLYCEMIA
- INCREASED OXYGEN DEMAND

- **4. IMPAIRED OXYGENATION AND VENTILATION**
  - EFFECTS OF ANESTHESIA
  - PAIN LIMITING CHEST EXPANSION
  - REDUCED MOBILITY

RESULT:

  - ATELECTASIS
  - HYPOXEMIA
  - RISK OF ACUTE PULMONARY FAILURE

- **5. ALTERED IMMUNE FUNCTION**
  - SURGICAL STRESS SUPPRESSES THE IMMUNE RESPONSE
  - **RESULT:**
    - INCREASED SUSCEPTIBILITY TO INFECTION
    - DELAYED WOUND HEALING
    - RISK OF SEPSIS

4. EXTENSIVE BURNS

- **DEFINITION**
  - EXTENSIVE BURNS ARE **LARGE SURFACE-AREA OR DEEP TISSUE INJURIES** (USUALLY >20–30% TOTAL BODY SURFACE AREA IN ADULTS) THAT CAUSE **SEVERE FLUID LOSS, INFLAMMATORY RESPONSE, AND MULTI-SYSTEM DYSFUNCTION**, LEADING TO A **LIFE-THREATENING ACUTE BIOLOGIC CRISIS**.

- 1. CAPILLARY PERMEABILITY AND FLUID SHIFT
  - THERMAL INJURY DAMAGES CAPILLARY WALLS
  - PLASMA LEAKS INTO INTERSTITIAL SPACES (THIRD SPACING)

RESULT:

  - SEVERE HYPOVOLEMIA
  - ↓ CARDIAC OUTPUT
  - BURN SHOCK

2. INFLAMMATORY AND STRESS RESPONSE

- MASSIVE RELEASE OF CYTOKINES AND INFLAMMATORY MEDIATORS
- ACTIVATION OF THE SYMPATHETIC NERVOUS SYSTEM

RESULT:

- TACHYCARDIA
- INCREASED METABOLIC RATE
- INCREASED OXYGEN CONSUMPTION

3. LOSS OF SKIN BARRIER

- DESTRUCTION OF PROTECTIVE SKIN LAYER

RESULT:

- HIGH RISK OF INFECTION
- EXCESSIVE EVAPORATIVE WATER AND HEAT LOSS
- HYPOTHERMIA

- **4. IMPAIRED OXYGENATION**
  - INHALATION INJURY
  - CARBON MONOXIDE POISONING
  - PULMONARY EDEMA

RESULT:

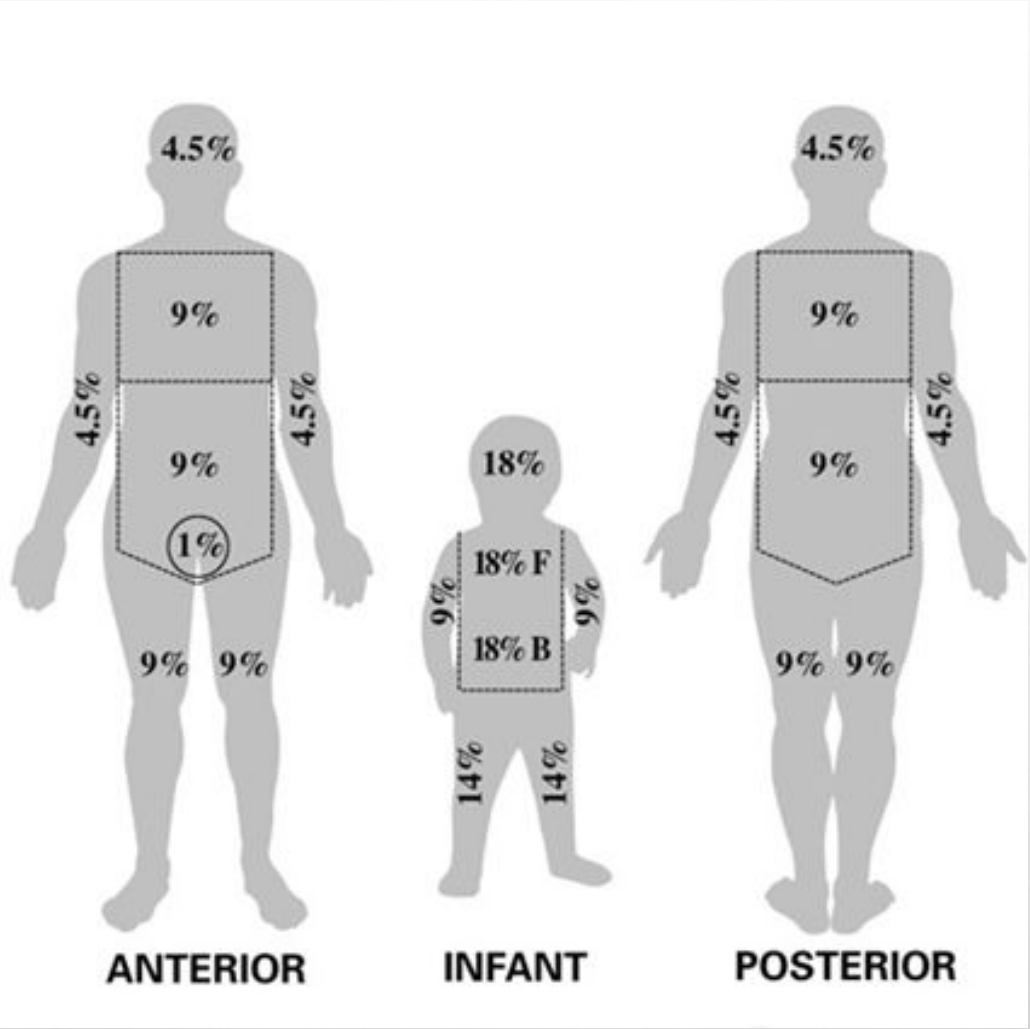
- HYPOXEMIA
- RISK OF ACUTE PULMONARY FAILURE

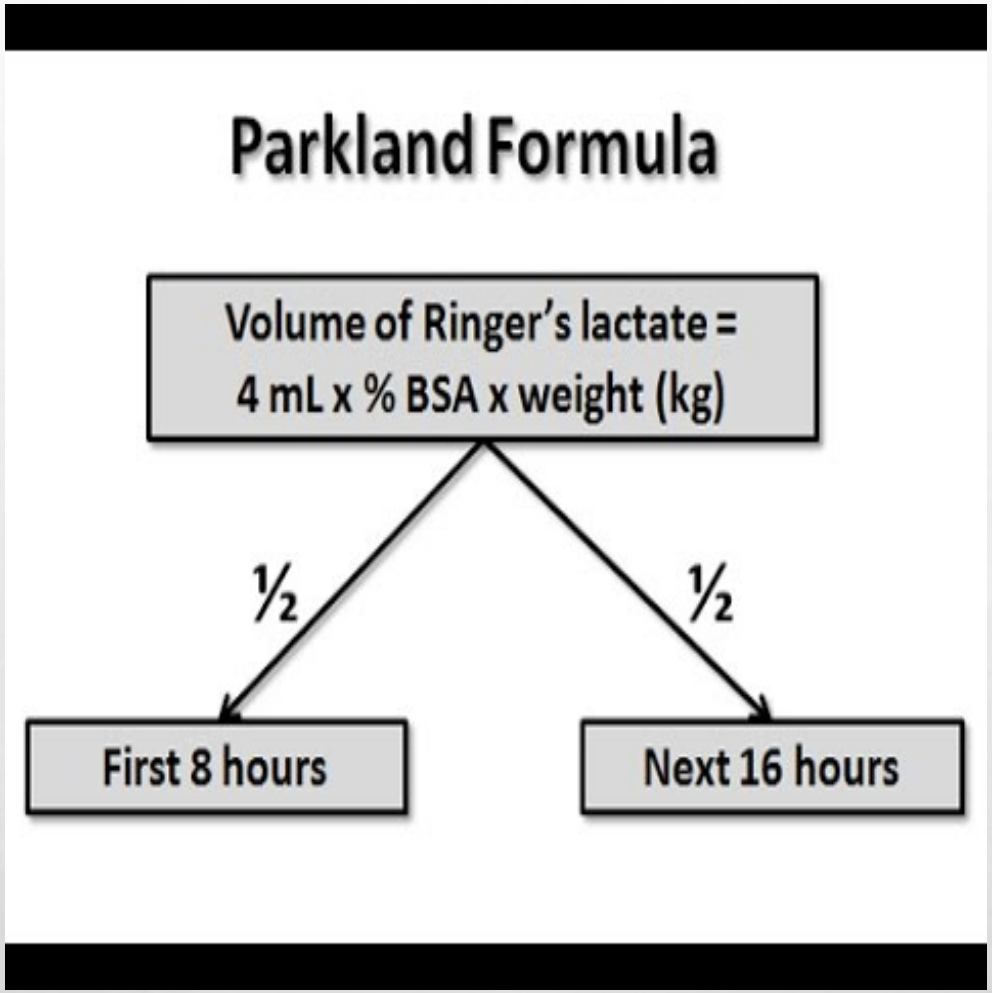
5. METABOLIC AND ELECTROLYTE IMBALANCE

- CELLULAR DESTRUCTION RELEASES POTASSIUM
- FLUID SHIFTS ALTER SODIUM LEVELS

RESULT:

- HYPERKALEMIA (RISK OF ARRHYTHMIAS)
- METABOLIC ACIDOSIS


4.5%
9%
4.5%
4.5%
9%
1%
9%
9%
ANTERIOR
18%
18% F
9%
9%
18% B
14%
14%
INFANT
4.5%
9%
4.5%
4.5%
9%
9%
9%
POSTERIOR


Parkland Formula
Volume of Ringer's lactate =
4 mL x % BSA x weight (kg)
½
½
First 8 hours
Next 16 hours

5. POISONING

- **DEFINITION**
  - POISONING IS THE **EXPOSURE TO TOXIC SUBSTANCES** (CHEMICAL, DRUG, BIOLOGICAL, OR ENVIRONMENTAL) THAT **INTERFERE WITH NORMAL CELLULAR AND ORGAN FUNCTION**, LEADING TO **ACUTE, POTENTIALLY LIFE-THREATENING SYSTEMIC EFFECTS**.

- **1. ABSORPTION OF THE TOXIN**
  - ENTRY VIA INGESTION, INHALATION, INJECTION, OR SKIN ABSORPTION

RESULT:

- RAPID ENTRY INTO THE BLOODSTREAM
- DISTRIBUTION TO TARGET ORGANS

2. CELLULAR AND ENZYMATIC DISRUPTION

- TOXINS INHIBIT ENZYMES OR DAMAGE CELL MEMBRANES
- INTERFERE WITH ATP PRODUCTION OR NEUROTRANSMISSION

RESULT:

- CELLULAR HYPOXIA
- METABOLIC ACIDOSIS
- ORGAN DYSFUNCTION

EXAMPLES:

- CYANIDE → BLOCKS CELLULAR RESPIRATION
- ORGANOPHOSPHATES → INHIBIT ACETYLCHOLINESTERASE

3. RESPIRATORY DEPRESSION OR FAILURE

- SUPPRESSION OF THE RESPIRATORY CENTER
- AIRWAY COMPROMISE

RESULT:

- HYPOVENTILATION
- HYPOXEMIA
- HYPERCAPNIA

EXAMPLES:

- OPIOID OVERDOSE
- SEDATIVE POISONING

4. CARDIOVASCULAR TOXICITY

- ALTERED MYOCARDIAL CONDUCTION OR CONTRACTILITY

RESULT:

- ARRHYTHMIAS
- HYPOTENSION
- CARDIAC ARREST

EXAMPLES:

- TRICYCLIC ANTIDEPRESSANTS
- BETA-BLOCKERS

5. ORGAN-SPECIFIC TOXICITY

- LIVER, KIDNEYS, AND BRAIN ARE MOST COMMONLY AFFECTED

RESULT:

- ACUTE LIVER FAILURE (E.G., ACETAMINOPHEN)
- ACUTE KIDNEY INJURY (E.G., ETHYLENE GLYCOL)
- CNS DEPRESSION OR SEIZURES

- **CLINICAL MANIFESTATIONS**
  - ALTERED LEVEL OF CONSCIOUSNESS
  - VOMITING, DIARRHEA
  - DYSPNEA OR CYANOSIS
  - ABNORMAL HEART RHYTHMS
  - SEIZURES
  - METABOLIC ACIDOSIS

- **NURSING IMPLICATIONS**
  - RAPID ABC ASSESSMENT
  - IDENTIFY SUBSTANCE, DOSE, AND TIME OF EXPOSURE
  - PREVENT FURTHER ABSORPTION (REMOVE CONTAMINATED CLOTHING)
  - MONITOR VITAL SIGNS AND NEUROLOGICAL STATUS
  - PREPARE FOR AIRWAY MANAGEMENT
  - ADMINISTER ANTIDOTES AS ORDERED

6. EMERGING ILLNESSES (SARS, MERS-COV, EBOLA)

- **DEFINITION**
  - EMERGING ILLNESSES ARE **NEW OR RE-EMERGING INFECTIOUS DISEASES** THAT CAUSE **SEVERE SYSTEMIC INFECTION, RAPID CLINICAL DETERIORATION, HIGH TRANSMISSIBILITY, AND HIGH MORTALITY**, OFTEN LEADING TO **ACUTE BIOLOGIC CRISIS**.

1. VIRAL ENTRY AND RAPID REPLICATION

- VIRUS ENTERS THROUGH RESPIRATORY TRACT OR MUCOUS MEMBRANES
- RAPID REPLICATION WITHIN HOST CELLS

RESULT:

- DIRECT CELLULAR DAMAGE
- HIGH VIRAL LOAD

- **2. SEVERE INFLAMMATORY AND IMMUNE RESPONSE**
  - EXCESSIVE RELEASE OF CYTOKINES (“CYTOKINE STORM”)

RESULT:

  - WIDESPREAD TISSUE INFLAMMATION
  - INCREASED CAPILLARY PERMEABILITY
  - HYPOTENSION AND SHOCK

3. ACUTE PULMONARY INVOLVEMENT

- ALVEOLAR DAMAGE AND FLUID ACCUMULATION

RESULT:

- IMPAIRED GAS EXCHANGE
- HYPOXEMIA
- ACUTE RESPIRATORY FAILURE / ARDS

- **4. MULTI-ORGAN DYSFUNCTION**
  - REDUCED TISSUE PERFUSION
  - DIRECT VIRAL TOXICITY
  - **RESULT:**
    - RENAL FAILURE
    - LIVER DYSFUNCTION
    - COAGULOPATHY

5. HIGH TRANSMISSION AND INFECTION RISK

- SPREAD VIA DROPLETS, CONTACT, OR BODY FLUIDS

RESULT:

- HEALTHCARE-ASSOCIATED TRANSMISSION
- NEED FOR STRICT INFECTION CONTROL

- **CLINICAL MANIFESTATIONS**
  - HIGH FEVER
  - SEVERE FATIGUE AND MYALGIA
  - RESPIRATORY DISTRESS
  - HYPOTENSION
  - BLEEDING (EBOLA)
  - MULTI-ORGAN FAILURE

NURSING IMPLICATIONS

- STRICT INFECTION PREVENTION AND ISOLATION
- CONTINUOUS MONITORING OF RESPIRATORY AND HEMODYNAMIC STATUS
- SUPPORTIVE CARE (OXYGEN, FLUIDS)
- USE OF PPE
- PSYCHOLOGICAL SUPPORT FOR PATIENTS AND FAMILIES

7. MULTIPLE INJURIES (POLYTRAUMA)

- **DEFINITION**
  - MULTIPLE INJURIES (POLYTRAUMA) REFER TO **SIMULTANEOUS INJURIES AFFECTING TWO OR MORE BODY SYSTEMS**, RESULTING IN **SEVERE PHYSIOLOGIC STRESS, RAPID DETERIORATION, AND A HIGH RISK OF SHOCK, RESPIRATORY FAILURE, AND MULTI-ORGAN DYSFUNCTION**.

1. COMBINED TISSUE DAMAGE

- BLUNT OR PENETRATING TRAUMA TO MULTIPLE ORGANS
- DISRUPTION OF NORMAL ANATOMY AND PHYSIOLOGY

RESULT:

- SEVERE PAIN
- INFLAMMATORY RESPONSE
- INCREASED METABOLIC DEMAND

2. HEMORRHAGE AND FLUID LOSS

- INTERNAL AND/OR EXTERNAL BLEEDING
- CAPILLARY LEAKAGE DUE TO TRAUMA

RESULT:

- HYPOVOLEMIA
- ↓ CARDIAC OUTPUT
- HYPOVOLEMIC SHOCK

- **3. IMPAIRED OXYGENATION**
  - CHEST TRAUMA (PULMONARY CONTUSION, PNEUMOTHORAX)
  - AIRWAY OBSTRUCTION OR ASPIRATION

RESULT:

  - HYPOXEMIA
  - ACUTE PULMONARY FAILURE

4. SYSTEMIC INFLAMMATORY RESPONSE (SIRS)

- MASSIVE CYTOKINE RELEASE FOLLOWING INJURY

RESULT:

- VASODILATION
- CAPILLARY LEAKAGE
- HYPOTENSION
- RISK OF SEPTIC-LIKE SHOCK

5. COAGULOPATHY

- CONSUMPTION OF CLOTTING FACTORS
- DILUTION FROM FLUID RESUSCITATION

RESULT:

- UNCONTROLLED BLEEDING
- WORSENING SHOCK

(PART OF THE TRAUMA “LETHAL TRIAD”)

- **6. PROGRESSION TO MULTI-ORGAN DYSFUNCTION**
  - PERSISTENT HYPOXIA AND HYPOPERFUSION

RESULT:

- ACUTE KIDNEY INJURY
- ARDS
- CARDIAC FAILURE
- DEATH IF UNTREATED

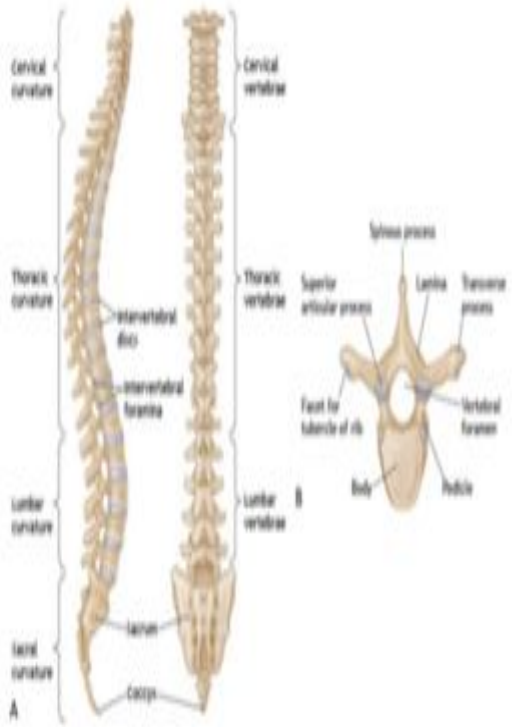
- **CLINICAL MANIFESTATIONS**
  - HYPOTENSION AND TACHYCARDIA
  - ALTERED LEVEL OF CONSCIOUSNESS
  - RESPIRATORY DISTRESS
  - MULTIPLE VISIBLE INJURIES
  - DECREASED URINE OUTPUT
  - SIGNS OF SHOCK

- **NURSING IMPLICATIONS**
  - RAPID PRIMARY SURVEY (ABCDE)
  - CONTROL BLEEDING AND IMMOBILIZE INJURIES
  - MONITOR VITAL SIGNS AND NEUROLOGICAL STATUS
  - MAINTAIN AIRWAY AND OXYGENATION
  - PREVENT HYPOTHERMIA
  - PREPARE FOR EMERGENCY PROCEDURES

Consider spine injury in all patients with multiple injuries.

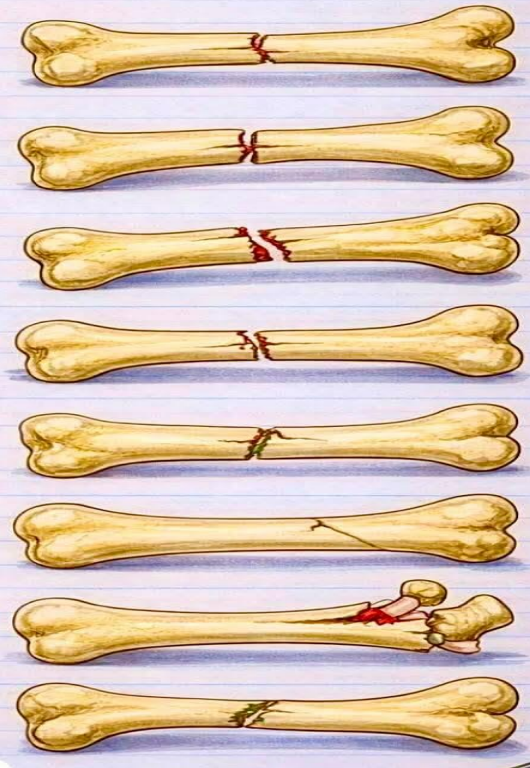
- Cervical region - 55%
- Thoracic region - 15%
- Thoracolumbar junction - 15%
- Lumbosacral area - 15%

Many patients with c-spine fractures have a second, noncontiguous vertebral column fracture.



TYPES OF FRACTURES

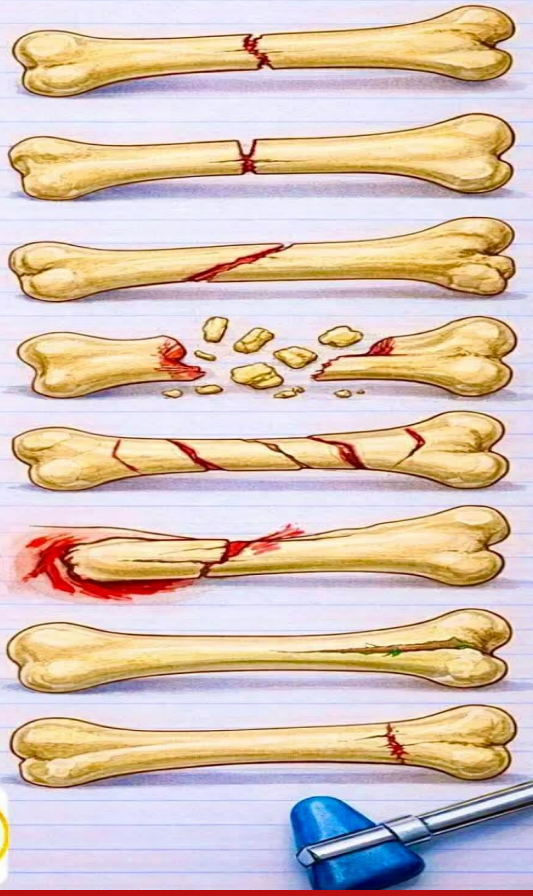
- **Closed Fracture:**
  - Broken bone that does not break the skin.
- **Open Fracture (Compound):**
  - Broken bone that pierces through the skin.
- **Displaced Fracture:**
  - Fractured bone ends are out of alignment.
- **Non-Displaced Fracture:**
  - Bone breaks but remains aligned.
- **Complete Fracture:**
  - Bone is broken completely through.
- **Incomplete Fracture:**
  - Bone cracks but does not break entirely.
- **Pathologic Fracture:**
  - Caused by disease or weakened bone.
- **Avulsion Fracture:**
  - Bone bends and cracks, common in children.



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• PYQs • Notes

TYPES OF FRACTURES

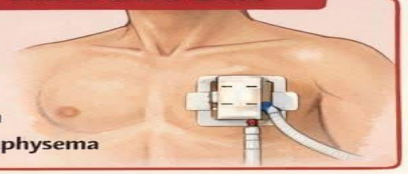
- **Stable Fracture:**
  - The bone breaks cleanly but stays aligned.
- **Transverse Fracture:**
  - Break is in a straight, horizontal line.
- **Oblique Fracture:**
  - Break has an angled pattern.
- **Comminuted Fracture:**
  - Bone shatters into 3 or more pieces.
- **Spiral Fracture:**
  - Caused by a twisting force, spirals around the bone.
- **Compound (Open) Fracture:**
  - Bone breaks through the skin, at risk of infection.
- **Hairline (Stress) Fracture:**
  - A small crack in the bone, often from overuse.
- **Greenstick Fracture:**
  - Bone bends and cracks, common in children.



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Chest Tube Drainage Care

Chest Tube Insertion Site

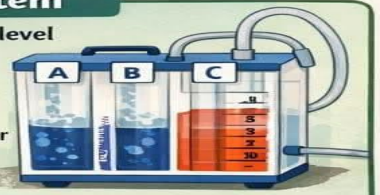


Assess for:

- Dressing intact
- No air leaks
- Signs of infection
- Subcutaneous emphysema

Drainage System

- Keep below chest level
- **Check for:**
- Water seal chamber Tidaling/bubbling
- Suution control chamber
- Collection chamber drainage
- Secure all connections



Patient Care

- Encourage Cough & Deep Breathing
- Repositioning
- Pain Management



Safety & Troubleshooting

- Keep clamps at bedside
- **If tube disconnects:**
  - Place end in sterile water
- **If tube dislodges:**
  - Cover with sterile dressing
  - Tape 3 sides
  - Call for help!



Monitoring & Assessment



- Respiratory status
- Lung sounds
- Vital signs
- Drainage output

Drainage Collection

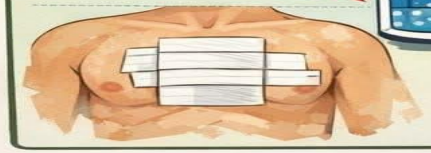
- Assess amount & color:
  - Serous
  - Sanguineous
  - Purulent
- Mark & date output levels
- No sudden increase



Air Leaks

- **Check for:**
- Continuous bubbling in water seal chamber
- Notify provider if air leak noted





BLOOD TRANSFUSION

What to Know for the NCLEX-RN Exam

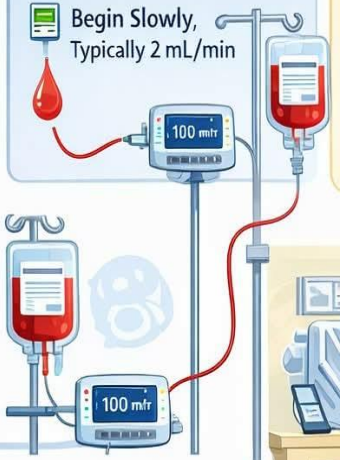
BEFORE TRANSFUSION:

- Verify Order
- Patient Consent
- Check Compatibility
- Assess Vitals
- Obtain Supplies

• Use 18-20 gauge IV catheter, normal saline (0.9% NaCl)

DURING TRANSFUSION

- Stay with Patient First 15 Minutes
- Monitor Vital Signs
- Begin Slowly, Typically 2 mL/min



MONITOR FOR COMPLICATIONS

- Fever & Chills
- Rash, Itching
- Back or Chest Pain
- Dyspnea (difficulty breathing)

Stop Transfusion If Reaction Occurs

KEEP V LINE OPEN WITH NORMAL SALINE, NOTIFY PROVIDER IMMEDIATELY.

AFTER TRANSFUSION:

- Vital Signs
- Document
- Dispose of Supplies

AFTER TRANSFISION:

- Vital Signs
- Document

• Vital Signs after 15 min, 30 min, hourly until complete, then post-transfusion




The moment you
GIVE UP
is the moment
you let someone
else win
KOBE BRYANT
BRYANT
8